JS

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JAVA SCRIPT

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WHAT IS JAVASCRIPT ?

- JavaScript is a programming language used to make websites interactive, dynamic, and responsive to user actions.

Think about an online shopping website:

You click “Add to Cart” 🛒 → item gets added.

You click “Buy Now” 👜 → checkout opens.

If You enter an “incorrect password” 🔑 → “Invalid password” appears.

You click “wishlist” ❤️ → product gets added to wishlist.

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CONNECTING JAVASCRIPT TO HTML

- **Option 1 : Inside HTML File**

  <script>\n console.log("Hello!");\n</script>

- **Option 2 : Separate file**

  <script src="script.js"></script>

Internal JS	External JS
Quick testing	Real projects
Small examples	Cleaner code
Inside HTML	Separate file
❌ Less maintainable	✅ Recommended

Key Takeaway

console.log() → print

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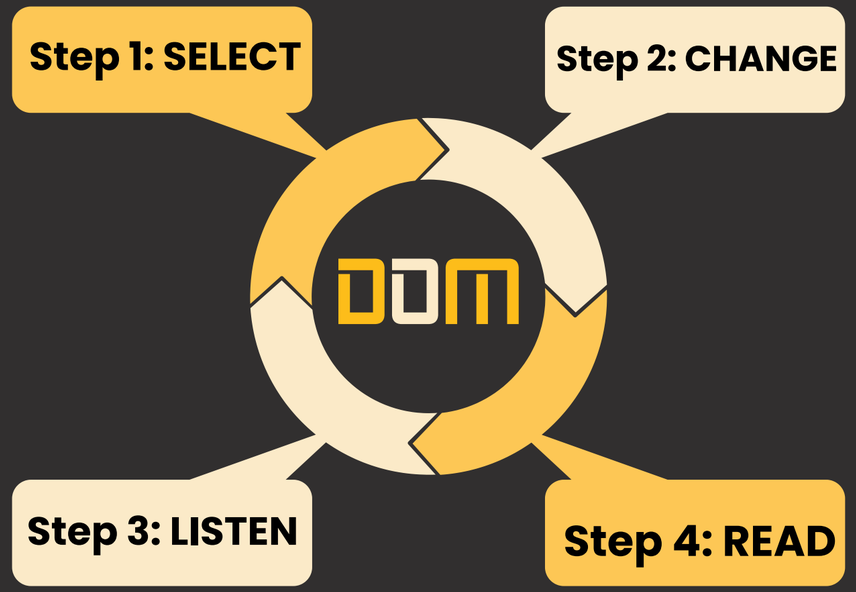
MEET THE DOM

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DOM - DOCUMENT OBJECT MODEL

- Think of the DOM as:

  JavaScript's remote control for your webpage 🎮



STEP 1 : SELECT

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Before JavaScript can change something...
It needs to find it!

Method	Example	Best for
getElementById()	#title	One specific element
querySelector()	.card	First matching element
querySelectorAll()	p	Multiple elements

Example

```
const title = document.getElementById("title");
```

Visual

HTML <h1 id="title">Hello</h1>

↓

getElementById("title")

↓

⚡ **JavaScript gets the element**

🧠 **Remember :**
→ ID
. → Class
tag → Element

STEP-2 : CHANGE

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Welcome to my site



Welcome to EdSpark! ⚡

JS

```
title.textContent = "Welcome to EdSpark!";
```

TWO IMPORTANT PROPERTIES

Property	What it does
textContent	Changes text
innerHTML	Changes HTML content

Examples

- element.textContent = "Hello!";
- element.innerHTML = "Hello!";

Mini Challenge !!

Change “Welcome” → “Let's Build!” when a button is clicked.

STEP 3 : LISTEN

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A webpage is full of events.

Click	click
Double click	dblclick
Type	input
Press a key	keydown
Submit form	submit
Move mouse	mouseover
Page loads	load

Basic pattern

```
element.addEventListener("click", () => {
  // What should happen?
});
```

Actually What Happens

👤 USER
↓
clicks
↓
🖱️ EVENT
↓
⚡ JAVASCRIPT
↓
💥 ACTION

Rememder :
Events are how JavaScript listens to the user.

STEP 4 : READ

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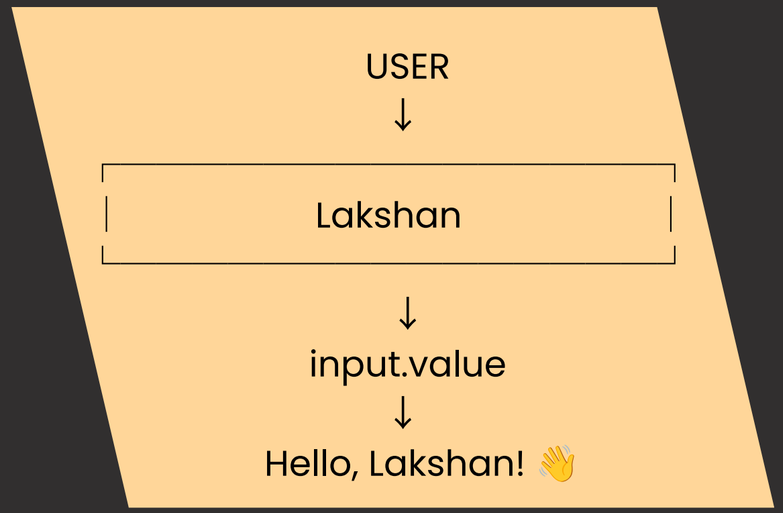
HTML :

```
<input id="name">
<button id="btn">Greet</button>
```

JAVASCRIPT :

```
const name = document.getElementById("name");
console.log(name.value);
```

value = What the user typed



Key Takeaway

Text input → .value

LOGIC IN ACTION: CONDITIONS & FUNCTIONS

- **Functions**: Reusable machines that run code only when you tell them to.
- **Conditions (if/else)**: Decision-making logic based on true/false checks.

```
function checkAge(age) {
  if (age >= 18) {
    console.log("Eligible to Vote! 🗳️");
  } else {
    console.log("Too young! 👶");
  }
}
checkAge(20);
```

Think of functions as recipes, and conditions as the decision points in that recipe

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THANK YOU

~EdSpark